
PLANTFORGE: A Procedural Control-Plant Corpus for In-Context System Identification

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Abstract

In-context system-identification (SysID) transformers are typically trained on a single private data-generation recipe — most commonly Wiener-Hammerstein-only white-noise data, regenerated independently by every paper in this line of work. We introduce PLANTFORGE, a released, static, reproducible corpus of synthetic control-plant trajectories organized along three axes jointly — nonlinearity *family*, excitation *class*, and sampling *rate* (exact zero-order-hold) — with per-instance Fisher-information identifiability annotations. Across 5 independently-seeded training runs, we show that a transformer trained on Wiener-Hammerstein-only data collapses off its training distribution (cross-family nMSE degradation of 93–118 $\times$), and that training on PLANTFORGE’s multi-axis corpus closes the rate and excitation gaps (to $\sim 1\text{--}2\times$) while only halving, not closing, the cross-family gap (93 $\times \rightarrow 14\times$). We then subject this result to three checks a released benchmark should survive but rarely does: (1) zero-shot transfer to two real measured plants (Silverbox, Cascaded Tanks), where corpus-trained models again outperform the Wiener-Hammerstein-only model, but a simple classical ARX baseline outperforms *both* trained transformers by 30–300 $\times$, tempering any claim of transformer superiority; (2) a within-cell statistical re-analysis showing that the corpus’s own identifiability annotations do *not* positively predict in-context prediction difficulty (median Spearman $r = -0.122$ across 40 cells), the opposite of our working hypothesis; and (3) full disclosure of a Simpson’s-paradox confound our first analysis of (2) missed, and how we caught and corrected it. We release the corpus (240,000 instances, CC BY 4.0), all generation/training/evaluation code (MIT), and every number in this paper with reproduction instructions.

1 Introduction

Model-agnostic, in-context system identification — training a sequence model to infer a plant’s dynamics from a short context of (u, y) pairs, with no gradient updates at test time — is an increasingly popular paradigm for control and mechatronics [Forgione et al., 2023, Piga et al., 2024]. The training recipe behind this line of work, however, is narrow and largely unreleased: each paper privately regenerates Wiener-Hammerstein-only white-noise data, the one family the incumbent generators support. No existing corpus varies *nonlinearity family*, *excitation class*, and *sampling rate* jointly, and none ships per-instance identifiability annotations that would let a user reason about *why* an instance is hard.

This paper makes two kinds of contribution. First, we release PLANTFORGE: a procedurally generated corpus spanning 5 nonlinearity families $\times$ 4 excitation classes $\times$ 3 sampling rates (240,000 instances total), with per-instance relative Cramér-Rao lower bounds (CRLB) and Fisher-information

matrix (FIM) condition numbers as identifiability annotations, and design invariants enforced by tests (Section 3). Second, and we think more valuably for the community, we report what happened when we subjected our own headline claim to increasingly rigorous scrutiny: multi-seed error bars, zero-shot real-plant validation, a classical baseline, and a statistical re-analysis of our identifiability hypothesis. Two of these checks produced results that *complicate* the paper’s original thesis rather than confirm it, and we report them in full rather than adjusting the narrative around them (Section 4).

Summary of findings.

- Multi-axis training on PLANTFORGE closes the rate and excitation transfer gaps that Wiener-Hammerstein-only training leaves open, but only halves the harder cross-family gap — reported here as an open challenge, not a solved problem (Section 4.1).
- Zero-shot on two real plants, a corpus-trained model beats a Wiener-Hammerstein-only model, but a simple per-window ARX fit beats *both* trained transformers by 30–300× under an identical, leakage-free evaluation protocol (Section 4.3).
- The corpus’s own identifiability annotations, hypothesized to predict in-context prediction difficulty, show a weak, robustly *negative* within-cell correlation with it once a between-cell confound in our first analysis is corrected (Section 4.4).

2 Related Work

Synthetic foundation models for dynamics. Ziegler et al. [2024] pretrain a transformer-based foundation model purely on synthetic dynamics sampled from a reproducing kernel Hilbert space, and show it transfers to real hardware after fine-tuning. This is generically capable but does not vary family/excitation/rate/identifiability jointly, and does not target control-plant nonlinearities specifically; we view it as the closest and most relevant prior synthetic-data approach, and the one whose scope PLANTFORGE is most easily confused with.

Scaling benchmarks for dynamical-system identification. DynaDojo [Bhamidipaty et al., 2023] is an extensible platform for benchmarking how SysID algorithms scale with training samples, system complexity, and target error, over a library of ~20 generic ODE/chaotic systems. It does not include control-specific nonlinearities (deadzone, saturation, hysteresis, drivetrain compliance) or an excitation/rate axis, and carries no identifiability annotations.

Real-measured benchmarks. The community’s default real-data benchmarks — Silverbox, Wiener-Hammerstein, Bouc-Wen, Cascaded Tanks [Wigren and Schoukens, 2013, Schoukens and Ljung, 2019] — are fixed, single-record, real-measured systems with no parameter ground truth. We evaluate zero-shot against two of them (Section 4.2); a classical baseline outperforming our trained transformers on those benchmarks (Section 4.3) is itself evidence for why a synthetic corpus with known parameters and controllable axes is a useful complement, not a replacement, for this real-measured suite.

In-context system identification. The training recipe we stress-test throughout this paper — Wiener-Hammerstein-only white-noise pretraining for an in-context SysID transformer — is the one used across the Forgone et al. [2023], Piga et al. [2024] line of work, each of which regenerates its own private WH-only data. PLANTFORGE’s motivation is to make this recipe’s narrowness measurable and to release a broader alternative, not to claim architectural novelty in the transformer itself: our InContextSysID model is a standard encoder-only transformer over (u, y) pairs.

3 The PLANTFORGE Corpus

Design axes. Every instance is generated along three axes jointly: **family** — `stribeck` (velocity-dependent friction, a state-nonlinearity), `backlash` (input deadzone), `saturate` (input clipping), `boucwen` (output hysteresis with memory), and `drivetrain` (two-inertia motor/gear/compliant-load with load Coulomb friction, the mechatronics staple) — plus a Wiener-Hammerstein (`wh`) family excluded from the corpus proper but retained for the failure experiment in Section 4.1; **excitation** — `prbs` (0.25 s physical hold), `multisine` (8 tones ≤ 1.85 Hz), `chirp` (0.05→3 Hz), and `closedloop`

(a true sequential PI-tracking loop against the plant stepper, so the same seed against a different plant instance produces different u — not a two-pass imitation of closed-loop data); and **rate** — 10/20/50 Hz, exact zero-order-hold discretizations of one shared continuous-time truth per instance.

Identifiability annotations. Each (instance, excitation, rate) combination carries a per-parameter relative Cramér-Rao lower bound and a Fisher-information-matrix $\log_{10}$ condition number, computed analytically from the same simulator that generated the trajectory.

Design invariants (tested). (1) (u, y) pairs are physically consistent with their stated parameters — no hidden output normalization; re-simulation from θ reproduces y to 10^{-6} . (2) Multi-rate instances are the *same* continuous-time plant: state-nonlinear families substep internally at ≤ 2 ms with zero-order-hold input, so dt and $dt/4$ agree at common sample instants to within 10^{-3} relative error (exact for LTI cores) — an axis we are not aware of any other released corpus generating exactly. (3) Closed-loop excitation is a genuine sequential loop, verified by checking that u depends on the specific plant instance under control. (4) Identifiability annotations are directionally physical: an excitation that never exits the backlash deadzone yields a relative CRLB on the deadzone width $\sim 5 \times 10^4$, versus ~ 0.3 for an excitation that does.

Release. 240,000 instances (5 families $\times$ 4 excitations $\times$ 3 rates $\times$ 4,000 instances/cell), 583 MB, released under CC BY 4.0 [Anonymous, 2026]. All generation, training, and evaluation code is released under MIT. Full datasheet (motivation, composition, collection process, uses, distribution, maintenance) accompanies the release.

4 Experiments

All models are an identical encoder-only transformer (InContextSysID: 5 layers, 8 heads, width 160) consuming 192 context samples and predicting 32 query-horizon samples with no ground-truth y visible in the query horizon. Every headline number in this section is the mean $\pm$ standard deviation over **5 independently-seeded training runs** (seeds 0–4, 10,000 steps each, fixed evaluation seeds 900–905 across every model for paired comparison). nMSE is always the ratio of batch-mean query-horizon squared error to batch-mean query-horizon signal power, never a mean of per-window ratios.

4.1 Headline: multi-axis training closes the rate/excitation gap, halves the family gap

We compare two training recipes: **wh_only**, trained exclusively on Wiener-Hammerstein white-noise data (the incumbent recipe), and **corpus**, trained on 4-of-5 PLANTFORGE families $\times$ {multisine, prbs} $\times$ {10, 50 Hz}, with backlash, chirp/closedloop, and 20 Hz held out.

Table 1: Synthetic transfer matrix, nMSE (mean $\pm$ std, $n = 5$ seeds). $\times$ ref is relative to that model’s own in-distribution reference cell.

held-out axis	wh_only nMSE	($\times$ ref)	corpus nMSE	($\times$ ref)
reference (train-like)	0.0040 $\pm$ 0.0004	1.0 $\times$	0.0215 $\pm$ 0.0017	1.0 $\times$
family backlash, dt=0.10	0.3880 $\pm$ 0.0322	97.7 $\times$	0.3107 $\pm$ 0.0080	14.5 $\times$
family backlash, dt=0.05	0.3678 $\pm$ 0.0408	92.6 $\times$	0.2899 $\pm$ 0.0095	13.5$\times$
family backlash, dt=0.02	0.4665 $\pm$ 0.0195	117.5 $\times$	0.3927 $\pm$ 0.0102	18.3 $\times$
excitation chirp	0.1533 $\pm$ 0.0645	38.6 $\times$	0.0383 $\pm$ 0.0143	1.8$\times$
excitation closedloop	0.0545 $\pm$ 0.0056	13.7 $\times$	0.0195 $\pm$ 0.0031	0.9$\times$
rate dt=0.05, stribeck	0.0341 $\pm$ 0.0039	8.6 $\times$	0.0215 $\pm$ 0.0017	1.0$\times$
rate dt=0.05, saturate	0.0141 $\pm$ 0.0016	3.5$\times$	0.0285 $\pm$ 0.0021	1.3 $\times$
rate dt=0.05, boucwen	0.0916 $\pm$ 0.0177	23.1 $\times$	0.0178 $\pm$ 0.0014	0.8$\times$
rate dt=0.05, drivetrain	0.2180 $\pm$ 0.0156	54.9 $\times$	0.1722 $\pm$ 0.0111	8.0$\times$

Table 1 and Figure 1 summarize the result. Multi-axis training **solves the rate and excitation axes for most families**: the WH-only model’s 8–39 $\times$ degradation on interpolated rate and held-out excitation collapses to ~ 1 –2 $\times$ under corpus training. It **halves but does not close the cross-family gap** (93–118 $\times$ $\rightarrow$ 14–18 $\times$, still an order of magnitude off reference) — we report this as PLANTFORGE’s open challenge track, not a solved problem. One cell, *saturate* at the held-out rate, is won by

wh_only (non-overlapping error bars the other direction): “corpus training helps” is not a universal law across every cell, and we report the exception rather than omit it. Non-overlapping ± 1 standard deviation across 11 simultaneous cell comparisons is a descriptive heuristic, not a multiplicity-corrected test; at $n = 5$ seeds this is a conservative bar relative to the standard error of the mean, and the largest-margin cells (below) clear it comfortably.

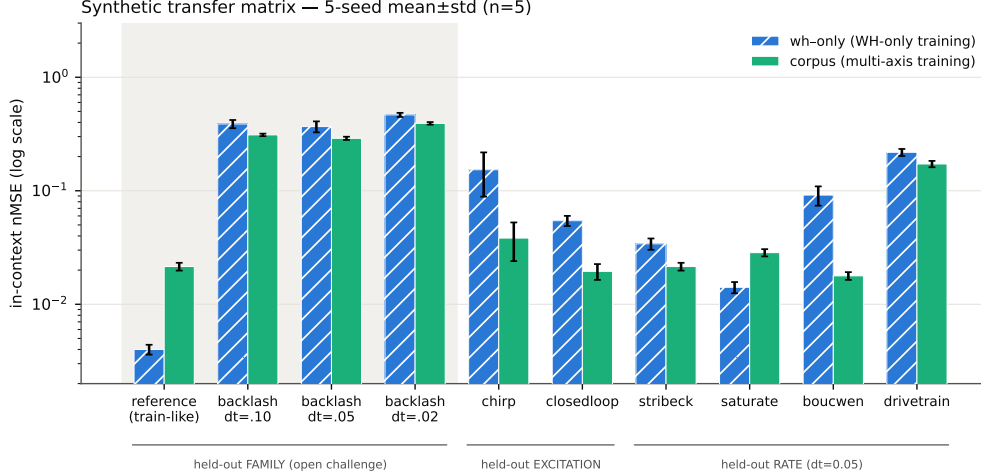


Figure 1: Synthetic transfer matrix, wh_only vs. corpus, 5-seed mean $\pm$ std, log scale.

4.2 Zero-shot transfer to real measured plants

Both checkpoints, trained only on synthetic PLANTFORGE data, are evaluated zero-shot (no fine-tuning) on real recordings from nonlinearbenchmark.org [Wigren and Schoukens, 2013, Schoukens and Ljung, 2019], obtained via the nonlinear_benchmarks package. Real sample rates do not align with PLANTFORGE’s three trained rates, so we handle each dataset on its own terms: **Silverbox** (native $dt \approx 1.64$ ms) is decimated $12\times$ (anti-aliased) to $dt \approx 19.7$ ms, landing near the finest trained rate; **Cascaded Tanks** (native $dt = 4.0$ s) is already $40\times$ coarser than the coarsest trained rate and is evaluated at its native rate as an explicit extrapolation; **Wiener-Hammerstein Process Noise** is excluded — its ~ 1.5 s total record duration is shorter than one 224-sample context window at any trained rate, so it cannot supply a single in-context evaluation window; **Bouc-Wen** is excluded, as it is not exposed by nonlinear_benchmarks’s public loader API.

Table 2: Zero-shot real-plant nMSE. wh_only/corpus: mean $\pm$ std, 5 seeds, fixed windows. ARX: single deterministic run, ≤ 8 non-overlapping windows (Section 4.3).

dataset	wh_only	corpus	ARX
Silverbox (decimated $12\times$)	0.958 ± 0.217	0.331 ± 0.040	0.0028
Cascaded Tanks (native rate, extrapolation)	0.394 ± 0.081	0.084 ± 0.043	0.0075

Corpus training again beats WH-only training zero-shot on real data (most dramatically on Silverbox), consistent with Section 4.1. But Table 2 previews a result that complicates the story further: a trivial classical baseline beats *both* trained transformers by two to three orders of magnitude. We discuss why in Section 4.3.

4.3 Classical baselines temper the transformer story

We compare against ARX and degree-2 polynomial NARX, fit by ordinary/ridge least squares *per window*, under an evaluation protocol identical to the transformer’s: fit (or, for the transformer, condition) on the 192-sample context only, then free-run the 32-sample query given only u — neither method ever observes query-horizon ground-truth y . ARX order $k \in \{2, 4, 8\}$ is selected per window by one-step-ahead error on the last 32 context samples, never touching the query.

Result. Table 2 and Figure 2 show ARX beating both trained transformers on both real-plant benchmarks by 30–300 $\times$. This pattern recurs synthetically: ARX also beats both models on held-out excitation chirp (0.0039 vs. 0.0383/0.1533), held-out excitation closedloop (0.0036 vs. 0.0195/0.0545), and held-out rate drivetrain (0.0253 vs. 0.1722/0.2180), and roughly ties corpus on held-out family backlash at the coarsest rate (0.3116 vs. 0.3107).

Is this a fair comparison? We verified it is. ARX and the transformer consume identical windows, identical normalization, and an identical context/query split; the transformer’s own query-horizon input is zeroed exactly where ARX is forbidden from using ground truth, so neither method sees query y at fit/inference time. ARX’s per-window least-squares fit is not an unfair privilege — *adapting to the context* is precisely the in-context transformer’s own value proposition, executed via a frozen forward pass instead of an explicit fit. The real advantage ARX has on rate-shifted and real-plant cells is structural, not a protocol bug: a per-window linear refit at the test sampling rate is immune to a rate mismatch that a frozen transformer must instead extrapolate through, most severely on Cascaded Tanks’ 40 $\times$ extrapolation. We view this as a genuine limitation of the frozen in-context approach worth further study, not an artifact to explain away.

Degree-2 NARX diverges broadly in free-run (clipped-but-huge nMSE, 10^{10} – 10^{11} , on all but the real-plant Silverbox cell) — a legitimate finding about naive polynomial NARX instability under 32-step free simulation, and partly attributable to the $k = 8$ order candidate being numerically underdetermined at the order-selection stage (153 quadratic features vs. 152 fit samples). We report it rather than omit it, but do not treat NARX2’s numbers as comparable to ARX’s.

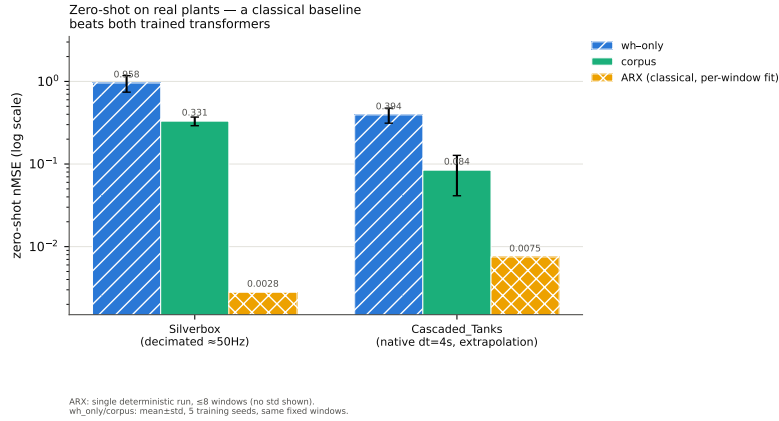


Figure 2: Zero-shot real-plant nMSE: wh_only, corpus, and ARX. Log scale.

4.4 Identifiability annotations do not predict prediction difficulty

PLANTFORGE’s per-instance rel-CRLB and FIM-condition annotations were hypothesized, at design time, to predict in-context prediction difficulty — intuitively, a poorly-identifiable instance should also be hard to predict. We tested this directly: per-instance query-horizon nMSE, scored on the corpus model (5 seeds, mean), against each instance’s identifiability annotation, on the 40 (family $\times$ excitation $\times$ rate) cells whose trajectory length exceeds one context window (10 Hz cells are excluded: 128 samples $<$ 224).

A confound we caught ourselves. Our first analysis pooled all 40,000 instances and correlated directly: Spearman $r = -0.088$ (max rel-CRLB) and $r = -0.316$ ($\log_{10}$ FIM condition), both statistically significant but with per-family breakdowns of *opposite sign* (backlash +0.294, drivetrain +0.091, vs. stribeck -0.081, saturate -0.148, boucwen -0.050). This sign heterogeneity is the fingerprint of a Simpson’s-paradox confound: excitation and family choice change both identifiability and model difficulty independently, so a pooled correlation conflates between-cell structure with the within-cell effect the hypothesis is actually about. We discarded the pooled analysis as the primary result and re-ran within-cell.

181 **Corrected analysis.** We compute Spearman r within each of the 40 cells separately, then aggregate.
182 We further apply two confound controls: (a) excluding each cell’s bottom decile of query-horizon
183 signal power (near-flat query horizons mechanically inflate nMSE, a live confound independent of
184 identifiability), and (b) restricting to cells with non-degenerate identifiability dynamic range (all
185 40/40 cells pass this filter).

Table 3: Within-cell Spearman r (max rel-CRLB, per-instance nMSE), aggregated over 40 cells, 5 seeds, 1,000 instances/cell (39,994/40,000 finite).

	median r	mean r	cells with $r > 0$
base	−0.122	−0.092	10/40
power-controlled (excl. bottom-decile power)	−0.117	−0.103	—
range-filtered (IQR > 10^{-6} , 40/40 cells kept)	−0.122	—	—

186 **Result: weak, robustly negative, not the hypothesized direction.** The within-cell correlation
187 is small and negative, and survives both confound controls without flipping sign (Table 3). Only
188 10/40 cells are positive, concentrated in backlash-multisine and backlash-closedloop (r up
189 to +0.327); every other family is negative on balance (Figure 3).

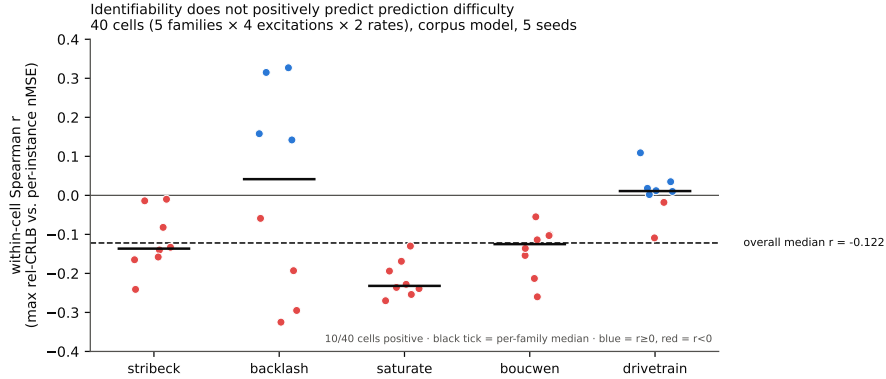


Figure 3: Within-cell Spearman r , all 40 cells, corpus model, 5 seeds. Ticks mark per-family medians.

190 **Interpretation.** We propose the negative sign reflects a real mechanistic decoupling rather than a
191 null result: rel-CRLB measures *parameter-recovery* difficulty, not *prediction* difficulty. A parameter
192 that is hard to identify from data is typically one with little influence on the output trajectory —
193 so an in-context predictor does not need to infer accurately what barely affects y in the first place.
194 Under this reading, the annotations remain valuable corpus metadata (e.g. for excitation-design
195 studies, where they are the intended quantity of interest) but are not, and should not be marketed as, a
196 prediction-difficulty predictor. A secondary methodological finding: naive quartile-binned *means* of
197 per-instance nMSE show a dramatic apparent upward trend ($1.5 \times 10^3 \rightarrow 1.6 \times 10^9$ across quartiles)
198 purely because per-instance nMSE has an extremely heavy right tail (near-zero query-horizon power
199 inflates the ratio); the quartile *medians* are flat (0.037, 0.037, 0.040, 0.021, Figure 4) — the mean-
200 based framing we initially reported for this experiment before catching the confound above was itself
201 a symptom of the same heavy-tail artifact, and we flag it explicitly so the failure mode is documented,
202 not just quietly fixed.

203 5 Discussion and Limitations

204 **What this paper does and does not claim.** We do *not* claim in-context transformers are the
205 strongest available method for real-plant system identification: Section 4.3’s ARX result argues
206 against that claim directly. We *do* claim that (a) training-distribution breadth matters — multi-axis
207 training measurably narrows, though does not close, the transfer gaps a narrow WH-only recipe leaves
208 open, and (b) a released, parameter-annotated, three-axis corpus is a more useful research artifact

Quartiles of max rel-CRLB $\rightarrow$ per-instance nMSE: mean is heavy-tail-dominated, median is flat ($n \approx 10,000/\text{bin}$)

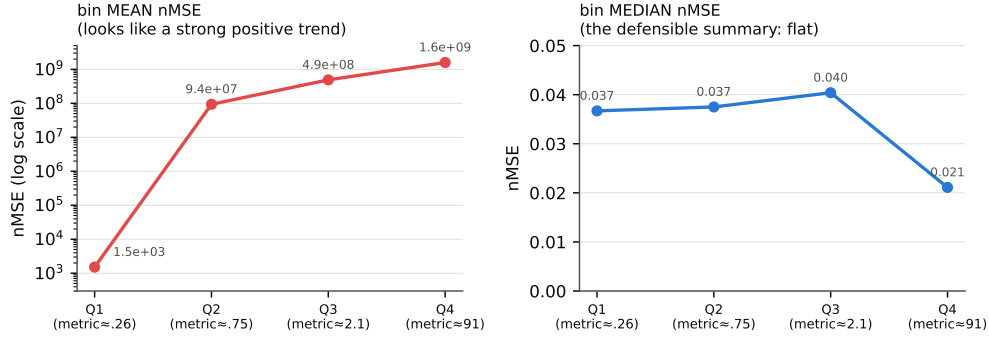


Figure 4: Quartiles of max rel-CRLB $\rightarrow$ per-instance nMSE: bin mean (left, heavy-tail-dominated) vs. bin median (right, flat).

for studying *why* than any of the alternatives in Section 2, independent of whether the transformer architecture itself is currently the best tool for the job.

Scope of real-plant validation. We validate zero-shot on 2 of the 4 real benchmarks named in our original design motivation (Silverbox, Cascaded Tanks); Wiener-Hammerstein Process Noise is excluded by record length and Bouc-Wen by loader availability, both explained in Section 4.2. Real-plant nMSE is computed over ≤ 8 non-overlapping windows per dataset; the transformer’s reported variance is over training seeds on this fixed window set, not over window sampling, and the ARX comparison point is a single deterministic run — the $30\text{--}300\times$ margins are far larger than plausible window-sampling noise, but this should be read as a first zero-shot probe, not an exhaustive real-plant evaluation.

Family coverage and held-out-family choice. We hold out a single family (backlash) throughout; we have not verified whether it is representative of the corpus’s hardest held-out-family case or an outlier, and leave a full leave-one-family-out sweep to future work. The corpus’s five families do not span every control nonlinearity of interest (no multi-input/multi-output plants, no time-varying parameters within a trajectory).

Architecture ablation: the family gap is capacity-invariant (single-seed check). All headline results (Sections 4.1–4.4) use one fixed transformer configuration (160 width, 5 layers, 1.58M params). We ran a lighter-weight, single-seed (seed 0, corpus recipe only) check across four additional variants spanning a $15\times$ parameter range, to test whether the held-out-family gap is an artifact of this specific size.

Table 4: Architecture ablation (single seed, corpus recipe): reference vs. held-out family backlash dt=0.05 nMSE.

variant	params	reference nMSE	family nMSE (dt=0.05)
narrow (d=80)	407k	0.0261	0.2939
default (d=160, L=5)	1.58M	0.0210	0.2865
shallow (L=2)	655k	0.0398	0.2835
deep (L=8)	2.51M	0.0151	0.2901
wide (d=320)	6.24M	0.0180	0.2941

The absolute family-transfer nMSE varies only 0.2835–0.2941 across this $15\times$ parameter span — a narrower spread than the default architecture’s own 5-seed standard deviation (± 0.0095 , Table 1) — while in-distribution reference nMSE improves monotonically with capacity on both the width and depth axes, exactly as expected. This is evidence, not merely an absence of counter-evidence, that the cross-family gap is a genuine open problem rather than an artifact of the paper’s specific model

size: scaling capacity roughly $4\times$ in either direction does not shrink it. A methodological note we flag explicitly: the $\times_{\text{ref}}$ ratio is misleading as a cross-architecture comparison metric on its own — deep appears to have the worst relative gap ($19.2\times$) and shallow the best ($7.1\times$) purely because their reference nMSE differs, not because their absolute family-transfer error does (0.2901 vs. 0.2835, within seed-to-seed noise) — the same class of trap as the mean-vs-median artifact in Section 4.4; absolute nMSE, not the ratio, should be read as primary. This check remains single-seed and does not cover context/query-length variation; a full multi-seed architecture sweep is left to future work.

On negative and inconvenient results. Two of this paper’s four experiments (Sections 4.3 and 4.4) produced results that complicate rather than support the paper’s motivating narrative. We report both in full, including the Simpson’s-paradox confound our own first identifiability analysis missed and how the corrected analysis was obtained, because we believe an honestly-reported null result and a caught methodological error are more useful to the community building on this corpus than a quieter framing would be.

6 Broader Impact

PLANTFORGE is a synthetic corpus of control-plant dynamics; we are not aware of misuse vectors specific to this release beyond those generic to open ML tooling. Positive impact: broadening the training distribution available for in-context SysID research may reduce duplicated private data-generation effort across the field (the “demand proof” motivating this corpus, Section 2) and, via the released identifiability annotations, support excitation-design research relevant to real experimental efficiency in control engineering.

7 Conclusion

We released PLANTFORGE, a three-axis synthetic control-plant corpus with identifiability annotations, and subjected our own headline claim to progressively harder scrutiny: multi-seed error bars, zero-shot real-plant transfer, a classical baseline, a corrected statistical re-analysis of our identifiability hypothesis, and a single-seed architecture ablation. The corpus’s central claim — broader training data narrows distribution-shift gaps for in-context SysID transformers — holds up under these checks, and the held-out-family gap specifically holds up as capacity-invariant across a $15\times$ parameter range, not an artifact of one model size. Two auxiliary claims we expected to hold did not: a classical baseline beats our trained transformers on several real and synthetic cells, and the identifiability annotations do not positively predict prediction difficulty. We report both, with the analysis that uncovered each, as part of the release.

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